

SHARATH B S, M.Sc., Ph.D.

Computational Chemistry | Bioinformatics | TechBio

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[Portfolio](#) · [LinkedIn](#)

PROFESSIONAL SUMMARY

Computational Research Scientist specializing in the intersection of AI-driven drug discovery and MLOps. Expert in developing end-to-end cheminformatics pipelines and Agentic AI/RAG workflows to automate scientific knowledge management and biological data analysis. Proficient in Python, HPC environments, and molecular modeling for rational drug design. Focused on building scalable, intelligent applications that transform complex biological and chemical data into actionable therapeutic insights.

CORE COMPETENCIES

Drug Discovery: Lead Optimization, Rational Peptide Design, Structure-Activity Relationship (SAR).

Computational Modeling: MD Simulations, Molecular Docking, Free-energy Calculations.

AI & Cheminformatics: Machine Learning (ML/DL), RDKit, MLOps, Agentic AI, LLMs.

Translational Science: CDISC SEND standards, Preclinical Toxicity Analysis, Systems Pharmacology.

Data Science: Multi-omics Analysis, Python Automation, HPC/SLURM Workflow Management.

PROFESSIONAL EXPERIENCE

Research Scientist

Dec 2025 – Present

[UR Advanced Therapeutics \(BioVaram\)](#) | Hyderabad, India

- Build computational pipelines for rational peptide design, and deployed ML models for peptide property prediction to guide candidate selection.
- Designed and optimized self-assembling and cell-penetrating peptides (CPPs) using advanced MD simulations to support biomaterial development.
- Performed multi-omics analysis for exosome characterization and biomarker prioritization

Postdoctoral Researcher

Feb 2024 – Mar 2025

[Pharmacoinformatics Research Group, Universität Wien](#) | Vienna, Austria

Project: Roche ROADS — In silico predictive & translational safety

- Preclinical data analysis using CDISC SEND format to support regulatory-ready translational safety assessments.
- Developed ML models to predict clinical outcomes and safety profiles from preclinical toxicity data, enabling data-driven decisions in drug development.
- Collaborated with cross-functional teams (Toxicologists) to map off-target effects and toxicity endpoints.

Postdoctoral Researcher

Aug 2021 – Dec 2023

[Soongsil University](#) | Seoul, South Korea

Project: In silico quorum-sensing inhibitor screening using cheminformatics tools (NRF, South Korea)

- Engineered end-to-end cheminformatics pipelines for high-throughput virtual screening, accelerating lead discovery from large compound libraries.
- Automated data extraction, processing, and proteochemometric modeling with reproducible Python workflows.
- Applied ligand- and structure-based CADD to identify and prioritize high-confidence drug candidates for quorum-sensing inhibitor screening.

EDUCATION

Ph.D. in Biotechnology

Feb 2019

[Kuvempu University](#) | Karnataka, India

Focus: Protein Interaction Networks & MD Simulations.

Thesis: Drug target identification through dynamics of protein interaction network analysis using VEGF-induced monocytes

M.Sc. in Biotechnology

Jun 2014

[Sahyadri Science College, Kuvempu University](#) | Karnataka, India

Focus: Nanoparticle Characterization & Toxicity.

Project: Phytosynthesis, characterization, and toxic effect of zinc nanoparticles on mitotic chromosomes of Allium cepa L.

B.Sc. in Chemistry, Biotechnology & Zoology

Jun 2013

[Sahyadri Science College, Kuvempu University](#) | Karnataka, India

CERTIFICATIONS

- **Machine Learning and Molecular AI** — Neovarsity.
- **Data Analysis for Genomics** — HarvardX /edX.
- **Version Control of a Python Project using Git** — Coursera.

TECHNICAL EXPERTISE

Molecular Modeling & Simulation:

Softwares: GROMACS, Amber, Martini (for coarse-grained MD), AutoDock, Vina/Smina, GNINA, DiffDock.

Methods: Classical MD, Steered MD, Umbrella Sampling, Replica Exchange, and membrane-protein MD, MM/GBSA, and MM/PBSA for binding free-energy. Molecular docking.

Visualization: PyMOL, ChimeraX, VMD.

AI, Cheminformatics & Data Science:

Cheminformatics: RDKit and Molecular descriptors/fingerprints (ECFP, MACCS etc), Pharmacophore & QSAR modeling.

MLOps libraries: Scikit-learn, TensorFlow, PyTorch, Pandas, Numpy, Matplotlib, Seaborn.

Next-Gen AI: Development of Agentic AI, LLMs, and RAG for automated scientific discovery.

Model Interpretation: SHAP for interpretability and dimensionality reduction tools like PCA, t-SNE, and UMAP.

Bioinformatics & Omics Pipelines:

NGS Ecosystem: Bioconductor and its core packages, Rstudio.

Primary Processing: SRA Toolkit, FastQC, MultiQC, Cutadapt, and Trim Galore.

Alignment & Quantification: STAR, Bowtie2, kallisto, RSEM, Samtools, and Subread.

Analysis & Networks: Differential expression via DESeq2, EdgeR; PPI/DDI network analysis, Gene Ontology/Pathway enrichment analysis using Cytoscape and its packages.

Programming & infrastructure:

Languages: Python and Bash scripting for automation.

Development Tools: Version control using Git/GitHub.

HPC: Management of large-scale simulations via SLURM and PBS schedulers.

Remote workflows: Workflow execution and management using SSH and SCP.

SELECTED PUBLICATIONS

Total: 27 publications · Citations: 314 · Full list: [Google Scholar](#)

- **Sharath B. S.**, Kandagalla S., Lee J.* Development of machine learning models based on molecular fingerprints for selectivity of novel small molecules against JAK2 protein. J. Comput. Chem. 2023 (IF 4.8). <https://doi.org/10.1002/jcc.27103>
- **Sharath B. S.**, Rimac H., Lee J.* In-silico screening of quorum sensing inhibitor candidates obtained by chemical similarity search. Molecules 2022 (IF 4.6). <https://doi.org/10.3390/molecules27154887>
- **Sharath B. S.**, Kandagalla S., Vikas H. M., Pavan Kumar G. S., Prabhakar B. T.*, Manjunatha H.* A systems biology approach to identify the key targets of curcumin and capsaicin that downregulate pro-inflammatory pathways in human monocytes. Comput. Biol. Chem. 2019 (IF 3.1). <https://doi.org/10.1016/j.compbiolchem.2019.107162>
- **Sharath B. S.**, Shivananda Kandagalla, Pavan Gollapalli, Bharath B R, Triveni H, Manjunatha Hanumanthappa*. Topology of protein-protein interaction network and edge reduction co-efficiency in VEGF signaling of breast cancer. NetMAHIB, 2017. IF-2.0 <https://doi.org/10.1007/s13721-017-0157-6>
- Kandagalla S.[†], **Sharath B. S.**[†], Rimac H.*, Grishina M., Manjunatha H.*, Potemkin V.* Computational insights into the binding mode of curcumin analogues against EP300 HAT domain as potent acetyltransferase inhibitors. J. Mol. Graph. Model. 2020 (†Equal contribution; IF 3.0). <https://doi.org/10.1016/j.jmgm.2020.107756>
- **Sharath B. S.**, Kandagalla S., Manjunatha H.* A network pharmacology approach to investigate the pharmacological effect of curcumin and capsaicin targets in cancer angiogenesis by module-based PPI network analysis. J. Proteins Proteomics 2019. <https://doi.org/10.1007/s42485-019-00012-y>

SELECTED CONFERENCES & WORKSHOPS

- International Conference on Biological Physics (ICBP 2023), Seoul, South Korea (Aug 14–18, 2023).
Presented title: “Computational Approach to Predict Off-target Binding Profile for Virtual Small Molecule Screening in Rational Drug Designing.”
- Lecture Notes in Computer Science (LNCS): Advances in Bioinformatics and Computational Biology (BSB 2020).
Presented title: “A systems biology driven approach to map the EP300 interactors using comprehensive protein interaction network.”
- 2023 Annual Conference of the Korean Society for Bioinformatics (BIOINFO 2023), Yeosu, South Korea (Nov 13–15, 2023).
- Hands-on Training in Genomics and Transcriptomics - jointly organized by The University of Trans-Disciplinary Health Sciences and Technology (TDU) and Bangalore Genomics Centre (BGC), Centre for Functional Genomics and Bioinformatics, Bengaluru, India.
- International Workshop on Computational Techniques for Wetlab Data Analysis (IWCWA), National College (Autonomous), Tiruchirappalli, India (Sep 22–24, 2015).